AppWorld
Mean ± SD

Method	Qwen2.5-7B		Llama-3.1-8B		Qwen3.8-27B	
	Stage 2	Stage 3	Stage 2	Stage 3	Stage 2	Stage 3
Frozen Base	0.2723 ±0.0316	0.2723 ±0.0316	0.2877 ±0.0285	0.2877 ±0.0285	0.3482 ±0.0337	0.3482 ±0.0337
AREW / T3	0.2877 ±0.0285	0.2913 ±0.0267	0.2913 ±0.0233	0.2954 ±0.0209	0.3764 ±0.0472	0.3968 ±0.0285
GiGPO	0.3036 ±0.0344	0.3154 ±0.0558	0.3241 ±0.0602	0.3482 ±0.0725	0.4087 ±0.0397	0.4323 ±0.0275
InfoPO	0.3518 ±0.0867	0.3882 ±0.0686	0.4046 ±0.0416	0.4282 ±0.0330	0.4809 ±0.0151	0.5082 ±0.0646
WorldEvolver	0.4650 ±0.0199	0.4487 ±0.0344	0.4532 ±0.0145	0.4568 ±0.0395	0.4836 ±0.0144	0.4805 ±0.0155
PoS / EnvProbe	0.4800 ±0.0158	0.4800 ±0.0158	0.5062 ±0.0289	0.5062 ±0.0289	**0.6430** ±0.0104	0.6430 ±0.0104
Step-DPO	0.4768 ±0.0200	0.5145 ±0.0588	0.4968 ±0.0438	0.5422 ±0.0794	0.5668 ±0.0853	0.6554 ±0.0772
Outcome-PG	0.2913 ±0.0233	0.3073 ±0.0308	0.2995 ±0.0162	0.3113 ±0.0277	0.4041 ±0.0644	0.4327 ±0.0510
Outcome-Value	0.2954 ±0.0304	0.3195 ±0.0521	0.3200 ±0.0517	0.3523 ±0.0663	0.4205 ±0.0445	0.4405 ±0.0452
Outcome-RLOO	0.2913 ±0.0233	0.3073 ±0.0308	0.3077 ±0.0304	0.3277 ±0.0663	0.4000 ±0.0698	0.4287 ±0.0568
Info-RL Neighbor	0.3195 ±0.0521	0.3523 ±0.0671	0.3564 ±0.0704	0.3964 ±0.0518	0.4446 ±0.0395	0.4687 ±0.0324
Unstratified	0.3200 ±0.0533	0.3641 ±0.0583	0.3641 ±0.0710	0.4082 ±0.0389	0.4650 ±0.0208	0.4768 ±0.0080
Unweighted	0.4394 ±0.0430	0.5895 ±0.1155	0.4650 ±0.0490	0.5374 ±0.0991	0.5423 ±0.0984	0.6257 ±0.0754
SPAR	**0.5284** ±0.0598	**0.5967** ±0.1090	**0.5091** ±0.0638	**0.6130** ±0.1241	0.6179 ±0.0753	**0.6906** ±0.0162